

SOLVING ONE OF NEW ENGLAND'S PARKING PROBLEMS

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Abstract—Given a $100 \times 200 \text{ ft}^2$ lot, the objective of this analysis is to determine a configuration of parking spaces in the lot that maximizes revenue from the lot. Valet parking vs self-parking was considered. Self-parking was preferred but an attendant would be needed to assure that parking fees were collected. Parking spaces angled from 45° to 90° were examined, in order to maximize the number of spaces in the lot. To minimize insurance claims uniformly, the access road width was computed depending on the angle of the space. The number of overall parking lot configurations was narrowed down to seven under the assumption that the fewer the number of turns in the access road of the lot, the greater the number of spaces which could be generated. Initially an analysis was done without considering entrances and exits or an attendant's station. A configuration was found which would hold 76 parking spaces. When the entrances and the booth were considered the configuration still generated more parking spaces (75) than any other configurations before these considerations. In our final solution, allowances for removal and temporary placement of snow, and spaces for lamp posts were made.

RESTATEMENT OF THE PROBLEM

Given a $100 \times 200 \text{ ft}^2$ corner lot in a New England town, design a parking lot configuration which maximizes the number of parking spaces, while minimizing difficult driving situations within the lot.

GENERAL ASSUMPTIONS

1. Cars will be self-parked. The cost of hiring an experienced driver to park cars will be too great to justify an increase in the total number of parking spaces in the lot.
2. The roads in the parking lot will all be one-way, in order to reduce both the width of the access roads and the total road area. In this way total parking area may be maximized.
3. Both the entrance and exit of the parking lot depend upon how the lot is situated on the street corner (see Fig. 1).

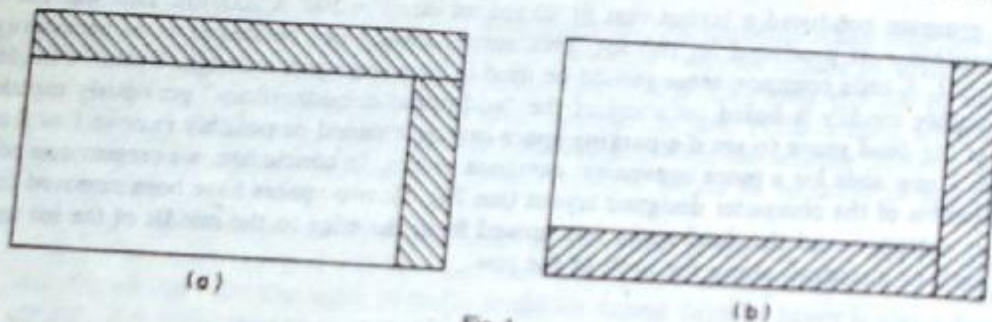


Fig. 1

4. Each parking space must have easy access to entrances and exits, i.e. no "first in, last out" scenarios.
5. It is necessary to have an attendant on the lot to insure parking fees are paid. Either the attendant can monitor parking meters on each space or collect fees directly. In either case it is necessary for the attendant to have a shelter (booth).
6. In the case of collecting fees directly, the booth must be located either between the entrance and exit, or if a ticket machine is used on the entrance, the booth must be located on the left of the exit.
7. In the analysis of maximizing the number of parking spaces, the position of both the entrance and exit and the booth need not be considered since a maximum of 6 spaces will be lost with their superposition.

8. Since the parking lot is in a New England town, the problem of snow and ice removal and temporary storage of snow in the lot must be considered.
9. The width of the parking space needed for drivers parking their own cars is 8 ft (note that this includes the 4 in. painted lines). The length of the parking space needed for drivers parking their own car is 17 ft.
10. The smallest outermost turn radius (i.e. the path of the front right bumper on a left-hand turn) of the average car is 21 ft. A 90° turn into a parking space requires a road width of 21 ft.
11. The minimal width of any roadway in the parking lot will be approx. 12 ft to allow safe driving in the lot.
12. Increasing the number of turns in the parking lot decreases the total number of parking spaces in the lot.
13. All parking spaces along any one access road are at the same angle in order to conserve road space and maximize parking area.
14. Snowfall uniformly affects any possible parking space configuration, so it need not be considered in designing a parking lot.

ANALYSIS AND MODEL DESIGN

In modeling the problem, we initially attempted to maximize the number of parking spots available while neglecting the positions of the entrance, exit and the booth.

First the angling of parking spaces was considered. With a fixed turning radius of 21 ft, the room necessary for pulling into a space at an arbitrary angle can be determined. In having the width of our access road dependent on the angle of parking spots, all parking spots can be entered with approximately the same amount of ease. A car turning into a space at 90° (right-angle parking), perpendicular to the access road, needs more room to turn in the access road than a car turning into a space angled at 45°. Thus insurance claims will be uniform in all configurations considered.

Right-angle parking is the densest means of parking, but in our model, it also requires the maximum width of access roads. As a result of this we considered other parking space angles, from 45° to 90°, in various combinations.

The outside of the car follows the path of the circumference of a circle with radius 21 ft. After proceeding along the circle to an arbitrary angle, θ , the car will be heading straight into a parking space at θ . The width of the access road, W_R , can be calculated by the following formula (see Fig. 2):

$$W_R = 21 - 13\cos(\theta).$$

The width and length of the parking space, W_S and L_S respectively, for an angle θ are also dependent on that angle, with formulas (see Fig. 3):

$$W_S = 8/\sin(\theta)$$

and

$$L_S = 17/\sin(\theta) + 8\cos(\theta).$$

Basically there are two ways of laying out a row of parked cars. A row may be parallel to a 200-ft side of the lot, or it may be parallel to a 100-ft side of the lot. Other layouts, such as ellipses and diagonal rows, are impractical because of the amount of wasted space.

Another useful tool in our analysis is the "dead length", L_D , the distance at the end of a row which is too small to park a car in. The formula for the dead length is (see Fig. 3):

$$L_D = 17\cos(\theta).$$

These tools were used in determining how many parking spaces there are in the parking lot for

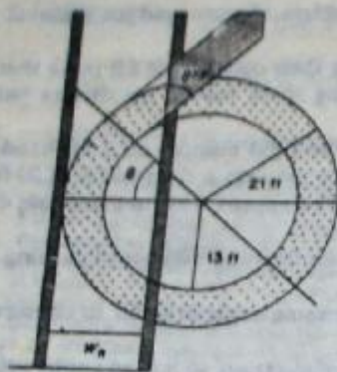


Fig. 2

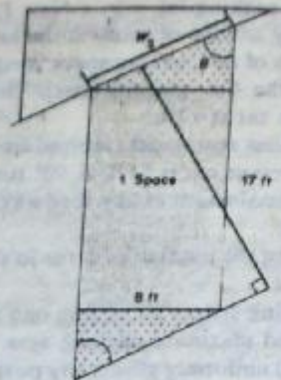


Fig. 3

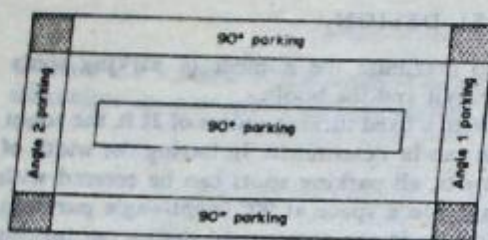


Fig. 4. Configuration 1.

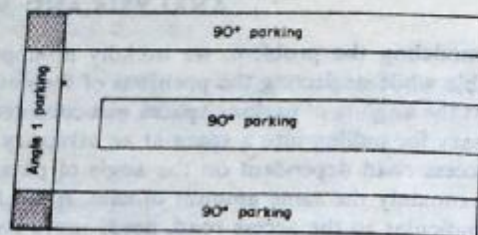


Fig. 5. Configuration 2.

a given configuration. It should be noted that, at most, two rows can have access to any one stretch of road.

Upon examining the different possible configurations of parking lots, we eliminated several configurations that were obviously poor choices due to the fact that they had many turns (see Assumption 12). We were then left with seven different configurations, most of which had rows parallel to the long side of the lot.

The first choice, Configuration 1, has three rows running parallel to the longer side of the lot, two of which are along the edge and the other in the middle of the lot, and two along the shorter edges (see Fig. 4). Since 90° parking is the densest parking possible and the three long rows could accommodate 90° parking, we arranged the three rows in such a manner. Configuration 2 is similar to Configuration 1 in that it has three rows of 90° parking; however it does not have parking along one of the shorter edges (see Fig. 5). Configuration 3 is the same as Configurations 1 and 2; it also has three long rows of 90° parking but no rows along the shorter edges (see Fig. 6). Configuration 4 is the same as Configuration 1 except that there is a double row in the middle and the long rows need not be 90° (see Fig. 7). Configuration 5 is the same as Configuration 2 with double rows in the middle, and again the long rows need not be 90° (see Fig. 8). Configuration 6 has four long

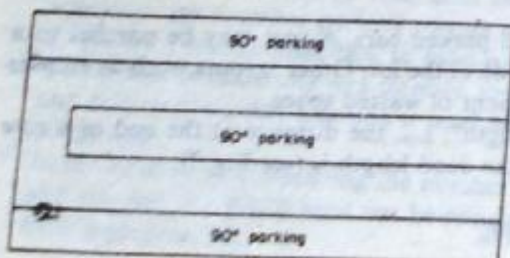


Fig. 6. Configuration 3.

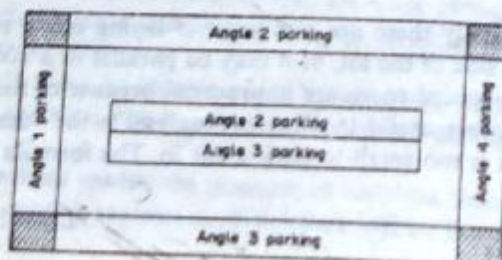


Fig. 7. Configuration 4.

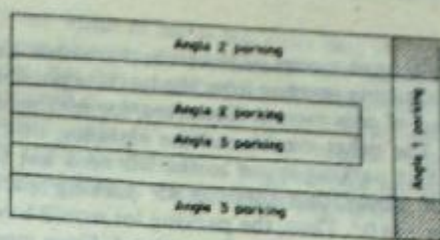


Fig. 8. Configuration 5.

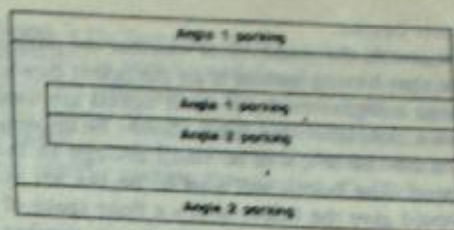


Fig. 9. Configuration 6.

rows, two of which are in the middle (see Fig. 9). Finally, Configuration 7 has all its rows parallel to the shorter side, having three double rows in the middle and at least one, possibly two, rows along the edges (see Fig. 10).

For each individual configuration, still ignoring the entrance, exit and the booth, we found the maximum number of parking spaces by checking all possible combinations of angles from 45° to 90° in increments of 1° . For each straight stretch of the access road all parking spaces along it have the same angle (see Assumption 13). To find the number of spaces in any particular row, we took the length of the row and subtracted L_0 from it and then divided that by W_1 and took the greatest integer, and this gave us the number of spaces in that row. See the Appendix for the FORTRAN program used for these particular calculations.

Having found the optimal set of angles for each configuration that maximize the number of parking spaces and the maximum number of spaces, we list our results in Table 1.

Clearly, Configuration 5 yields the most parking spaces in our simplified model. The closest of the other models, Configuration 3, has 3 fewer spaces. Since Configuration 5 has both an entrance and an exit already, the number of parking spaces that have to be removed is at most 2 or 3 in order to accommodate a parking lot booth. Therefore, Configuration 5 has more parking spaces even with the booth than all the others without one, hence it is the optimal solution.

Now we consider the placement of the exit, the entrance and the booth, previously neglected. From the open end of the parking lot the entrance will be on the left and the exit will be on the right. The access road will go in a clockwise direction (this is not the normal flow of traffic around this type of parking lot). Entering the lot, there are parking spaces at 63° on both sides until reaching the turn. At the far end of the lot, the parking spaces are perpendicular to the road. Exiting the lot, the parking spaces on either side are at 45° (see Fig. 11).

The entrance to the parking lot depends on whether the lot resembles Fig. 1(a) or 1(b), where the shaded region represents the neighboring streets. If the lot resembles Fig. 1(a) then the exit will be closer to the street corner, and if the lot resembles Fig. 1(b), then the entrance will be closer to

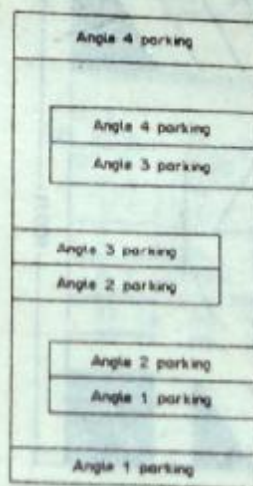


Fig. 10. Configuration 7.

Table 1					
	No. of spaces	Angle 1	Angle 2	Angle 3	Angle 4
Config. 1	71	90°	76°	NA	NA
Config. 2	72	76°	NA	NA	NA
Config. 3	73	NA	NA	NA	NA
Config. 4	72	90°	63°	45°	90°
Config. 5	76	90°	63°	45°	NA
Config. 6	72	63°	45°	NA	NA
Config. 7	72	90°	90°	90°	90°

the street corner.

Now comes the question of whether the parking lot will have meters or tickets and a booth. Let us note that having meters is an inefficient means of maximizing revenue from the lot. It is possible to charge a higher rate per car if tickets are used instead of coin meters. Therefore, for maximum revenue a ticket/booth system should be implemented. The ticket machine at the entrance, which must be on the left, will be small enough so that the first parking space to the left need not be eliminated. The booth will also be on the left at the exit, eliminating only one 45° parking space. This would give the ticket booth a floor space of at least 144 ft^2 . Thus, the parking lot would have 75 parking spaces, having the maximum number of parking spaces in Configuration 5. Since most of the parking spaces are at $<90^\circ$ and they all have ample turning radii, the average driver will have no trouble in maneuvering through the parking lot, thereby keeping the insurance claims to a minimum.

Note that lamp posts can be placed in "dead spaces", the spaces too small to park a car in, wherever appropriate. Or the dead spaces can be used in the winter months to temporarily store excess snow until it can be removed. Note also that snowfall would effect all configurations equally so it need not be considered in the analysis (see Assumption 14).

TESTING THE MODEL

The easiest means of testing the model is to check to see if the assumptions are fair. Checking the average width and length of cars revealed that nearly all passenger cars comfortably fit into an $8 \times 17 \text{ ft}^2$ parking space. We also observed that the average New England driver can easily complete a circle within a 42 ft diameter under all but the most severe conditions. Going through the final result and checking to see if a car can actually fit into all the parking spots with ease resulted in noticing that the first parking space on the left of the 45° parking spaces would be hard to get into, but that space can easily be converted into a 90° parking spot without any trouble.

STRENGTHS AND WEAKNESSES

Some of the weaknesses of our analysis lie in the fact that we may not have considered a configuration that might yield more parking spaces than Configuration 5, however if you accept our assumptions this is unlikely. Another weakness of our analysis is that we have quietly assumed that the front tires of a car are at the front bumper to simplify the analysis of the road widths for each angle. However, we have assumed that the width of a car is 8 ft in this analysis, whereas in

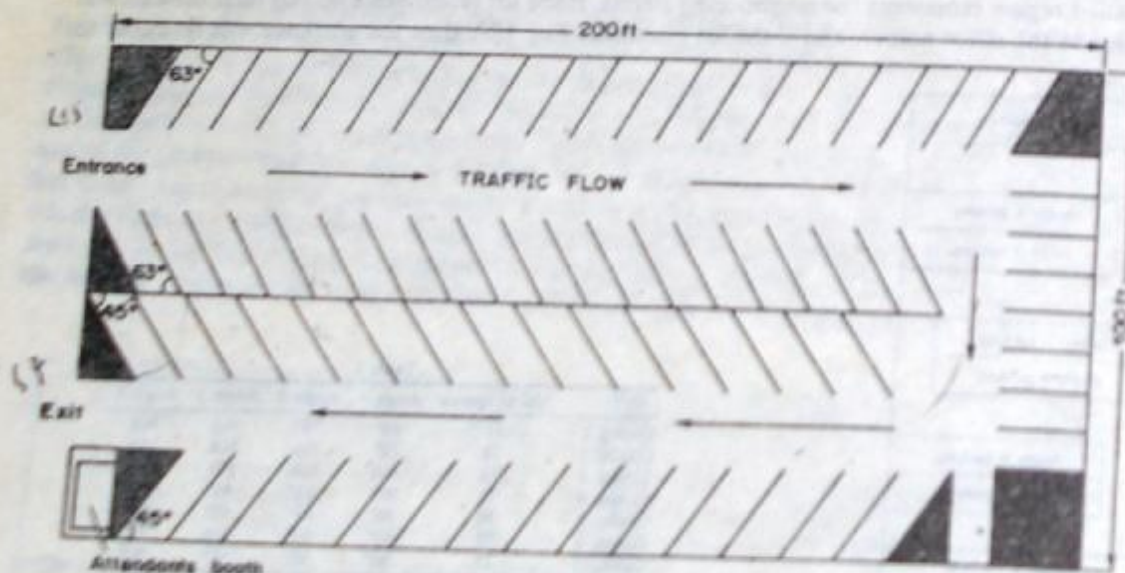


Fig. 11. Final solution—Configuration 5: 75 parking spaces.

actuality the width of car is less than that. A third weakness is the fact that we have not allowed any distance between the parking lot and the street (or sidewalk) to comply with local town or city ordinances, but this would affect all configurations. One of the strengths of the model is that we have optimized the set of angles of the parking spaces to the nearest degree, while the usual practice is to consider only 45°, 60° and 90° angles. Another strength is that by varying the road width with each angle we maintain a large enough turning radius, thereby making the number of accident claims independent of the design configuration.

APPENDIX -

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C      1987 MCM: PARKING LOT
C
C      THETA: ANGLE OF THE PARKING SPOT ( 90 DEG BEING PERPENDICULAR
C      TO THE ACCESS ROAD)
C      WS: DENOTES THE WIDTH OF THE PARKING SPOT, DEPENDENT ON THETA
C      LS: DENOTES THE LENGTH OF THE PARKING SPOT, ALSO DEPENDENT ON
C      THETA
C      WR: DENOTES THE WIDTH OF THE ACCESS ROAD IN ORDER FOR THE
C      "AVERAGE" DRIVER TO COMFORTABLY PULL INTO THE PARKING SPOT
C      LD: DENOTES "DEAD LENGTH", THE DISTANCE AT THE END OF A ROW WHICH
C      IS TOO SMALL TO PARK A CAR IN
C
C      REAL WS(50), THETA, LS(50), WR(50), LD(50), WTR(50)
C
C      X: DENOTES A DUMMY VARIABLE THAT STORES THE NUMBER OF SPACES
C      FOR EACH GIVEN CONFIGURATION AND SET OF THETAS
C      T: DENOTES ANOTHER DUMMY VARIABLE THAT KEEPS TRACK OF THE
C      GREATEST NUMBER OF SPACES FOR EACH OF THE SEVEN
C      DIFFERENT CONFIGURATIONS
C      CA(1,1): DENOTES THE MAXIMUM NUMBER OF PARKING SPOTS FOR A GIVEN
C      CONFIGURATION
C
C      INTEGER X, T, CA(10,10)
C
C      THE DO LOOP GENERATES ALL THE NECESSARY COMPONENTS IN ORDER
C      TO OPTIMIZE THE NUMBER OF SPACES IN EACH OF THE SEVEN
C      CONFIGURATIONS
C
C      DO 10 K = 45, 90
C      THETA = (3.1415*K) / 180
C      WS(K) = 8 / (SIN(THETA))
C      LS(K) = 17 * SIN(THETA) + (8*COS(THETA))
C      WR(K) = 21 - (13*COS(THETA))
C      LD(K) = 17 * COS(THETA)
C 10 CONTINUE
C      WRITE (6,20)
C 20 FORMAT (3X, '# OF SPACES', 3X, 'ANGLE 1', 3X, 'ANGLE 2', 3X,
C      'ANGLE 3', 3X, 'ANGLE 4')
C
C      -----CONFIGURATION ONE-----
C
C      T = 0
C      DO 40 I = 45, 90
C      DO 30 J = 45, 90
C      A = (100 - (2*17)) / WS(1)
C      B = (100 - (2*17)) / WS(J)
C      C = (200 - (LS(1) + LS(J))) / 8
C      D = (200 - (LS(1) + LS(J) + WR(1) + WR(J))) / 8
C      E = INT(A) + INT(B) + (2*INT(C)) + INT(D)
C      CA(1,1) = MAX0(X,T)
C      IF (X .GT. T) CA(1,2) = 1
C      IF (X .GT. T) CA(1,3) = J
    
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T = X
30 CONTINUE
40 CONTINUE
WRITE (6,50) CA(1,1), CA(1,2), CA(1,3)
50 FORMAT (110, 2X, 19, 2X, 18)

C
C
C
C
-----CONFIGURATION TWO-----
T = 0
DO 70 I = 45, 90
  A = (100 - (2*17)) / WS(1)
  B = (200 - LS(1)) / 8
  C = (200 - (LS(1) + WR(1))) / 8
  X = INT(A) + (2*INT(B)) + INT(C)
60 FORMAT (110, 1X, 110)
  CA(2,1) = MAX0(X,T)
  IF (X .GT. T) CA(2,2) = I
  T = X
70 CONTINUE
WRITE (6,60) CA(2,1), CA(2,2)

C
C
C
C
-----CONFIGURATION THREE-----
A = 200 / 8
CA(3,1) = (3*A) - 2
WRITE (6,80) CA(3,1)
80 FORMAT (110)

C
C
C
C
-----CONFIGURATION FOUR-----
T = 0
DO 120 I = 45, 90
  DO 110 J = 45, 90
    DO 100 K = 45, J
      DO 90 L = 45, I
        A = 2 * LS(J) + 2 * LS(K) + WR(J) + WR(K)
        IF (A .GT. 100.) GO TO 90
        B = (100 - LS(J) - LS(K)) / WS(1)
        C = (100 - LS(J) - LS(K)) / WS(L)
        D = (200 - LS(I) - LS(L)) / WS(J)
        E = (200 - LS(I) - LS(L)) / WS(K)
        F = (200 - LS(I) - LS(K) - MAX1(WR(I),12.) - LD(J)) / WS(J)
        G = (200 - LS(I) - LS(K) - MAX1(WR(I),12.) - LD(K)) / WS(K)
        X = INT(A) + INT(B) + INT(C) + INT(D) + INT(E) + INT(F)
        CA(4,1) = MAX0(X,T)
        IF (X .GT. T) CA(4,2) = I
        IF (X .GT. T) CA(4,3) = J
        IF (X .GT. T) CA(4,4) = K
        IF (X .GT. T) CA(4,5) = L
90 CONTINUE
100 CONTINUE
110 CONTINUE
120 CONTINUE
WRITE (6,130) CA(4,1), CA(4,2), CA(4,3), CA(4,4), CA(4,5)
130 FORMAT (110, 1X, 3110, 1X, 110)

C
C
C
C
-----CONFIGURATION FIVE-----
DO 160 I = 45, 90
  DO 150 J = 45, 90
    DO 140 K = 45, J
      A = 2 * LS(J) + 2 * LS(K) + WR(J) + WR(K)
      IF (A .GT. 100.) GO TO 140
      B = (100 - LS(J) - LS(K)) / WS(1)
      C = (100 - LS(J) - LS(K)) / WS(J)
      D = (200 - LS(I) - LD(K)) / WS(K)
      E = (200 - LS(I) - MAX1(WR(I),12.) - LD(J)) / WS(J)
      F = (200 - LS(I) - MAX1(WR(I),12.) - LD(K)) / WS(K)
      X = INT(A) + INT(B) + INT(C) + INT(D) + INT(E) + INT(F)
      CA(5,1) = MAX0(X,T)

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IF (X .GT. T) CA(5,2) = I
IF (X .GT. T) CA(5,3) = J
IF (X .GT. T) CA(5,4) = K
140 CONTINUE
150 CONTINUE
160 CONTINUE
WRITE (6,170) CA(5,1), CA(5,2), CA(5,3), CA(5,4)
170 FORMAT (I10, 1X, 3I10)
C
C
C -----CONFIGURATION SIX-----
DO 190 I = 45, 90
DO 180 J = 45, 1
A = 2 * LS(I) + 2 * LS(J) + WR(I) + WR(J)
IF (A .GT. 100.) GO TO 180
A = (200 - LD(I)) / WS(I)
B = (200 - LD(J)) / WS(J)
C = (200 - 12 - LD(I)) / WS(I)
D = (200 - 12 - LD(J)) / WS(J)
X = INT(A) + INT(B) + INT(C) + INT(D)
CA(6,1) = MAX0(X,T)
IF (X .GT. T) CA(6,2) = I
IF (X .GT. T) CA(6,3) = J
180 CONTINUE
190 CONTINUE
WRITE (6,50) CA(6,1), CA(6,2), CA(6,3)
C
C
C -----CONFIGURATION SEVEN-----
DO 250 I = 45, 90
DO 240 J = 45, 1
DO 230 K = 45, J
DO 220 L = 45, K
A = (100 - LD(I)) / WS(I)
B = (100 - LD(I) - 12) / WS(I)
C = (100 - 12 - LD(J)) / WS(J)
D = (100 - 12 - LD(K)) / WS(K)
E = (100 - 12 - LD(L)) / WS(L)
F = 2 * LS(I) + WR(I) + 2 * LS(J) + WR(J) + 2 * LS(K) +
WR(K) + 2 * LS(L) + WR(L)
IF (F .GT. 200.) GO TO 200
F = (100 - LD(L)) / WS(L)
GO TO 210
200 F = 0.
210 X = INT(A) + INT(B) + 2 * INT(C) + 2 * INT(D) + INT(E) +
INT(F)
CA(7,1) = MAX0(X,T)
IF (X .GT. T) CA(7,2) = I
IF (X .GT. T) CA(7,3) = J
IF (X .GT. T) CA(7,4) = K
IF (X .GT. T) CA(7,5) = L
220 CONTINUE
230 CONTINUE
240 CONTINUE
250 CONTINUE
WRITE (6,130) CA(7,1), CA(7,2), CA(7,3), CA(7,4), CA(7,5)
STOP
END

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